

Ruslan Klassen

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PROFESSIONAL SUMMARY

Machine Learning Scientist and Bioinformatics Ph.D. candidate with 5+ years of experience applying advanced machine learning and deep learning to large-scale biological datasets. Proven expertise in developing end-to-end, reproducible data pipelines and scalable predictive models using Python and PyTorch. Strong background in computational biology and life sciences, with a passion for driving actionable, data-driven insights in pharmaceutical research. Adept at cross-functional collaboration and translating complex analytical output into robust business solutions.

TECHNICAL SKILLS

Programming & Tools: Python, R, Bash, Git, SQL

Machine Learning & Deep Learning: PyTorch, scikit-learn, predictive modeling, statistical analysis

Data Visualization & Pipelines: Data cleaning, feature engineering, vision/data pipelines, Matplotlib, Seaborn

Software & Infrastructure: FastAPI, Flask, Docker, HTML/CSS, Linux environment

PROFESSIONAL EXPERIENCE

Product Manager & Senior Bioinformatician

GeneCore Facility | Institute of Biotechnology, Czech Academy of Sciences | 2026 – Present

- **Product Ownership:** Champion the end-to-end lifecycle of genomic data products, successfully delivering complex next-generation sequencing (NGS) analysis pipelines to client laboratories and external stakeholders.
- **Agile Project Management:** Lead cross-functional team workflows using the Kanban framework, effectively managing project backlogs, optimizing analytical turnaround times, and driving continuous process improvement.
- **Stakeholder Consultation:** Serve as the primary technical and product liaison, translating intricate multi-omic datasets into clear, actionable, and structured insights for collaborative research teams and clients.

Bioinformatics & Data Scientist

Institute of Biotechnology, Czech Academy of Sciences | 2020 – Present

- **Technical Leadership:** Acted as the technical lead for multiple laboratory projects, effectively managing project timelines, coordinating developer resources, and aligning technical outputs with the Principal Investigator's strategic research goals.
- **Machine Learning Deployment:** Delivered end-to-end machine learning and deep learning solutions using PyTorch and scikit-learn for advanced pattern detection, successfully moving models from initial research and prototyping to reliable, production-ready execution.

- **Data Pipeline Engineering:** Architected and maintained robust, scalable data pipelines utilizing Python and Bash, implementing core software engineering principles to guarantee absolute reproducibility and operational efficiency across projects.
- **Data Preprocessing & Robustness:** Engineered specialized workflows to clean and process noisy, imperfect, and high-dimensional biological datasets, expertly mitigating heavy class imbalances and domain shifts to ensure the mathematical validity of downstream models.
- **Cross-Functional Collaboration:** Partnered closely with multidisciplinary research teams, interpreting complex biological data outputs to provide clear, actionable insights and strategic recommendations to non-technical stakeholders.
- **Workflow Standardization:** Contributed directly to the design and implementation of internal best practices, significantly modernizing tooling, computing infrastructure, and analytical workflow standardization across the institute.

Laboratory Assistant & Assistant Teacher

University of Chemistry and Technology, Prague | 2020 – 2024

- **Operations & Compliance:** Supervised daily laboratory operations and enforced strict compliance with institutional safety, biochemical waste disposal, and quality control standards.
- **Academic Mentorship:** Mentored and guided undergraduate and postgraduate students, sharpening the ability to translate highly complex biochemical algorithms and bioinformatics concepts into accessible, structured instruction.
- **Performance Evaluation:** Evaluated student research initiatives, technical coursework, and laboratory examinations, providing structured, data-driven feedback to accelerate academic and technical skill development.
- **Faculty Collaboration:** Partnered closely with senior academic staff, researchers, and faculty members in a collaborative, team-oriented environment to optimize lab curricula and advance departmental research objectives.

Bioprocess Production Associate (GMP)

Lonza | Kouřim, Czech Republic | 2019 – 2019

- **GMP Compliance:** Executed daily operational workflows on the manufacturing floor within a strict, regulated **cGMP (Current Good Manufacturing Practice)** environment, prioritizing quality control, safety protocols, and operational excellence.
- **Sterilization & Upkeep:** Prepared, sterilized, and validated critical production line instrumentation utilizing advanced autoclaving procedures and sterile handling techniques to prevent cross-contamination.
- **Process Integrity:** Maintained meticulous batch documentation and equipment logs, ensuring total traceability and alignment with stringent corporate and international regulatory audits.

Research Scientist

Institute of Experimental Botany, Czech Academy of Sciences | 2017 – 2020

- **Advanced Microscopy:** Conducted high-resolution cellular imaging and spatial-temporal analysis utilizing cutting-edge techniques, including **spinning disk, confocal microscopy, FRET, and FRAP**.
- **Experimental Breeding:** Oversaw the complex lifecycle management, cultivation, and strategic breeding of experimental plant models to isolate specific genetic phenotypes.

- **Data Synthesis:** Developed structured, high-volume imaging data pipelines to capture, quantify, and analyze physiological variations for downstream peer-reviewed research.
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PROJECTS

Primer Design Web Application

Tech Stack: Python, Flask, Docker | [Live Application](#) | [2nd version](#)

- **Workflow Automation:** Identified a critical operational bottleneck in the standard RT-qPCR laboratory workflow and engineered a custom web application to automate primer design, significantly accelerating experiment planning and eliminating manual analytical errors for researchers.
 - **Backend Architecture:** Developed and optimized the end-to-end web application logic utilizing Flask, incorporating robust input validation protocols, data sanitization, and user-focused features to support highly complex molecular workflows.
 - **DevOps & Cloud Deployment:** Containerized the entire application infrastructure using Docker to guarantee absolute cross-platform portability and successfully deployed the environment into a scalable cloud infrastructure.
 - **Full-Lifecycle Ownership:** Executed the entire software development lifecycle (SDLC) independently, demonstrating total engineering ownership from initial user-need assessment through prototyping, rigorous testing, and final production deployment.
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EDUCATION

Ph.D. in Bioinformatics (*ongoing*)

University of Chemistry and Technology, Prague

M.Sc. (Ing.) in Biochemistry

University of Chemistry and Technology, Prague

B.Sc. (Bc.) in Biochemistry and Biotechnology

University of Chemistry and Technology, Prague

SOFT SKILLS

- Analytical thinking and structured problem-solving
 - Cross-functional collaboration
 - Communication of complex data to non-technical audiences
 - Adaptability and continuous learning
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LANGUAGES

- Russian: Native
- Czech: Fluent
- English: Proficient
- German: Basic